

Week 1 - Lab Session 1: Creating Flowcharts with Lucidchart

Objective:

- Learn to create clear and logical flowcharts using **Lucidchart**.
 - Apply problem-solving skills to design simple processes and workflows.
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Instructions:

1. Open **Lucidchart** on your desktop.
 - If you don't have an account, sign up for a free account.
 - Familiarize yourself with basic tools: shapes (oval for start/end, rectangles for processes, diamonds for decisions), connectors, and text boxes.
 2. Complete the following three activities by designing flowcharts in **Lucidchart**.
 3. Ensure your flowcharts follow these principles:
 - Begin and end with **oval shapes** (Start and End).
 - Use **rectangles** for steps/processes and **diamonds** for decision points.
 - Connect shapes with arrows to show the flow of the process.
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Activities:

Activity 1: Make a Sandwich

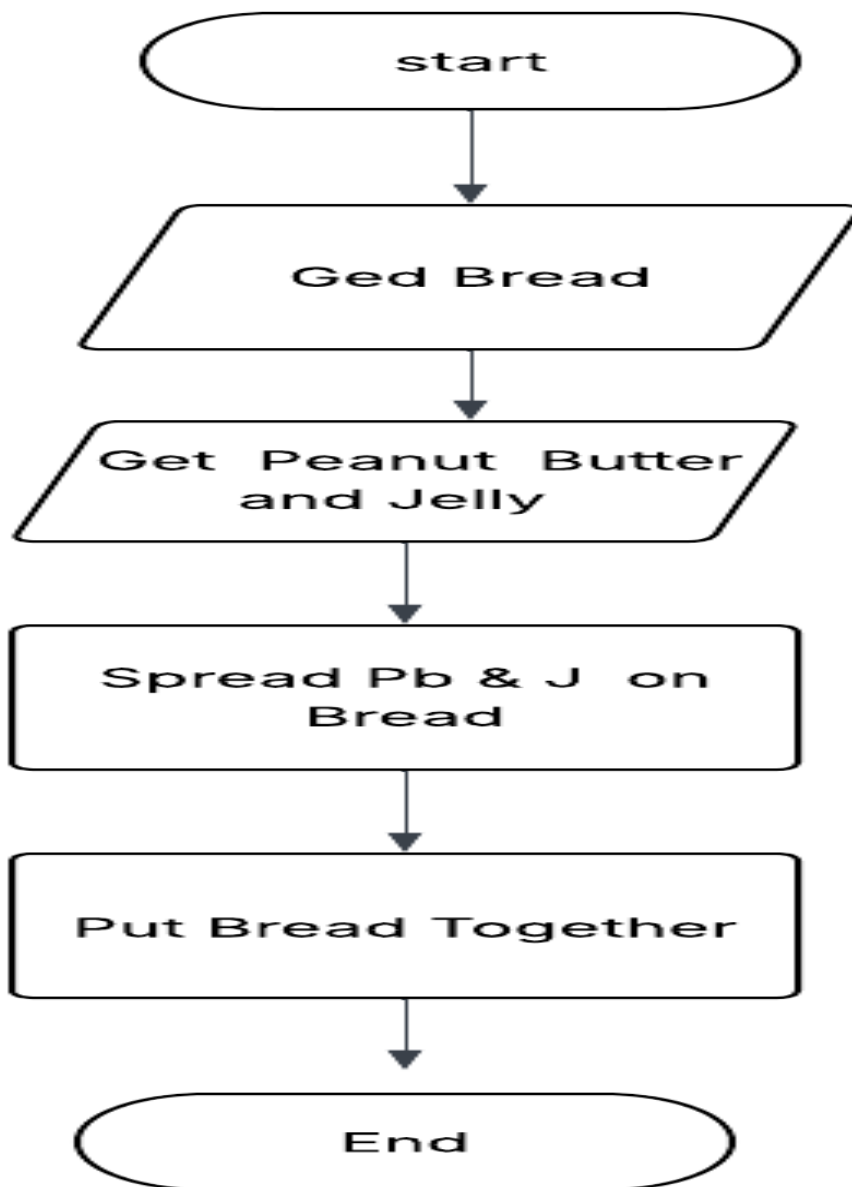
Task: Design a flowchart that outlines the process of making a peanut butter and jelly sandwich.

Flowchart Outline:

- **Start** → Get Bread → Get Peanut Butter and Jelly → Spread PB&J on Bread → Put Bread Together → **End**

Example Design:

- Start with an **oval** labeled "Start".
- Use rectangles for actions like "Get Bread" and "Spread PB&J on Bread".
- End with an **oval** labeled "End".
- Connect the shapes with arrows to show the flow of the steps.



Activity 2: Determine if a Number is Odd or Even

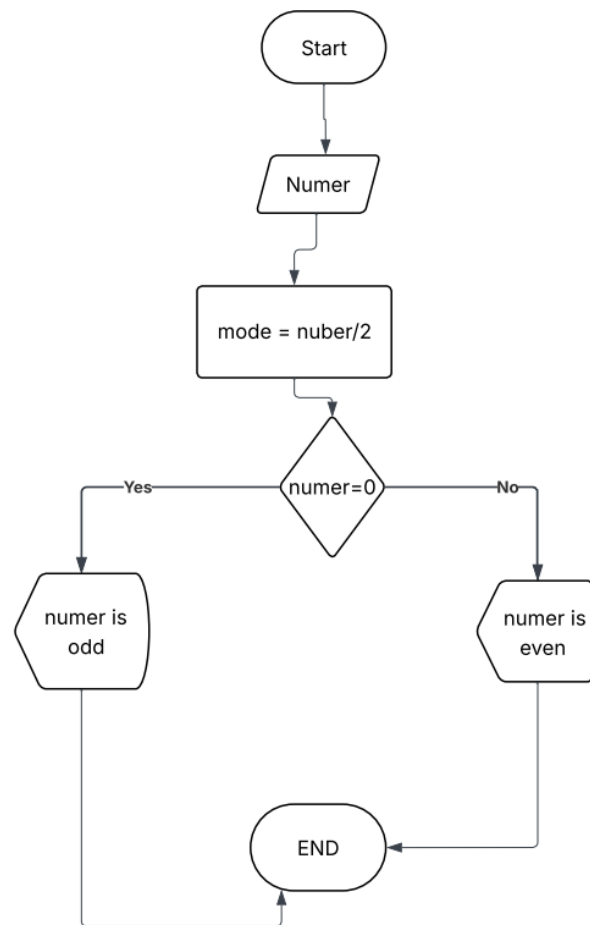
Task: Create a flowchart to decide if a given number is odd or even.

Flowchart Outline:

- **Start** → Input Number → Is Number mod 2 = 0?
 - **Yes** → Display "Even" → **End**
 - **No** → Display "Odd" → **End**

Design Tip:

- Use a **diamond shape** for the decision point "Is Number mod 2 = 0?"
- Label the arrows from the diamond as **Yes** and **No** for clarity.



Activity 3: Count to 10

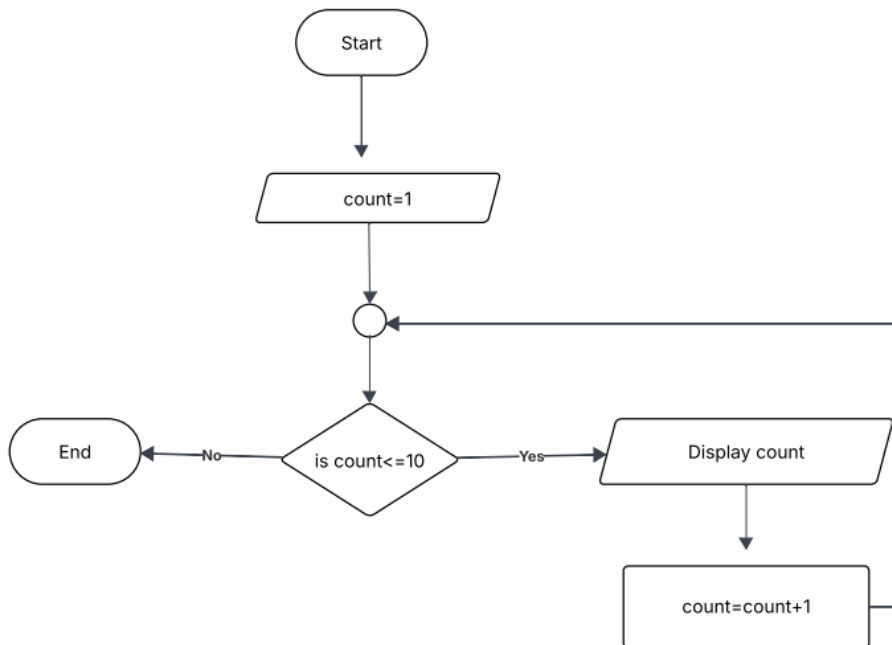
Task: Design a flowchart for a process that counts from 1 to 10.

Flowchart Outline:

- **Start** → Set Counter to 1 → Display Counter → Add 1 to Counter → Is Counter ≤ 10 ?
 - **Yes** → Go back to "Display Counter"
 - **No** → **End**

Design Tip:

- Use a **looping flow** for this activity. Connect the decision "Is Counter ≤ 10 ?" back to "Display Counter" for the **Yes** condition.
- Clearly label the flow between each step.



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Submission Requirements:

1. Save your flowcharts in Lucidchart.
2. Export each flowchart as a PDF or PNG file.
3. Submit your files by reply to the Lab in CANVAS

Key Notes:

- Flowcharts must be **clear** and **accurate**.
 - Use proper shapes (oval, rectangle, diamond) and **label all steps** and decisions.
 - Creativity is encouraged! Feel free to add visuals or details to your flowcharts.
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